



PARSHVANATH CHARITABLE TRUST'S

A. P. SHAH INSTITUTE OF TECHNOLOGY

Department of Information Technology

(NBA Accredited)



Semester: VII

Academic Year: 2026-27

Class / Branch: BE IT-C

Subject: Recent Open Source Lab

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EXPERIMENT NO. 05

Aim: To study open-source database management tools like Cassandra, MySQL, and understand the steps for contributing in Cassandra.

For Cassandra :

```
apsit@apsit-HP-280-G3-MT:~/Desktop/Shreya$ docker start cassandra-1
cassandra-1
apsit@apsit-HP-280-G3-MT:~/Desktop/Shreya$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
9097b8915ac6   cassandra:latest "docker-entrypoint.s..." 4 days ago    Up 18 seconds  7000-7001/tcp, 7199/tcp, 9042/tcp, 9160/tcp  cassandra-1
apsit@apsit-HP-280-G3-MT:~/Desktop/Shreya$ docker exec -it cassandra-1 nodetool status
Exception in thread "main" java.lang.IllegalArgumentException: Server is not initialized yet, cannot run nodetool.
    at org.apache.cassandra.tools.NodeTool.err(NodeTool.java:328)
    at org.apache.cassandra.tools.NodeTool.execute(NodeTool.java:291)
    at org.apache.cassandra.tools.NodeTool.main(NodeTool.java:87)
apsit@apsit-HP-280-G3-MT:~/Desktop/Shreya$ docker exec -it cassandra-1 bash -c 'cqlsh'

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.
```

**Installing cassandra through docker start cassandra-1 , after this execute
docker exec -it cassandra-1 bash -c 'cqlsh'**



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```
apsit@apsit-HP-280-G3-MT:~/Desktop/Shreya$ docker exec -it cassandra-1 bash -c 'cqlsh'

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.8 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE use_cassandra with replication = {'class':'SimpleStrategy' , 'replication_factor': 1};
cqlsh> CREATE TABLE use_cassandra.employee ( employeeId int PRIMARY KEY, name text, salary text );
cqlsh> INSERT INTO use_cassandra.employee(employeeId,name,salary) VALUES (23104134, 'Aditi', '75000');
cqlsh> INSERT INTO use_cassandra.employee(employeeId,name,salary) VALUES (23104124, 'Siddhi', '60000');
...
cqlsh> INSERT INTO use_cassandra.employee(employeeId,name,salary) VALUES (23104124, 'Siddhi', '60000');
cqlsh> INSERT INTO use_cassandra.employee(employeeId,name,salary) VALUES (23104148, 'Shreya', '80000');
cqlsh> SELECT *FROM use_cassandra.employee;

employeeid | name | salary
-----+-----+-----
23104148 | Shreya | 80000
23104134 | Aditi | 75000
23104124 | Siddhi | 60000

(3 rows)
cqlsh>
```

After execution insert the data like CREATE, INSERT and to run the table – SELECT *FROM use_cassandra.employee;



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2) Mysql

```
apsit@apsit-HP-280-G3-MT:~/Desktop/Shreya$ sudo systemctl start mysql
[sudo] password for apsit:
apsit@apsit-HP-280-G3-MT:~/Desktop/Shreya$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-08-03 12:49:28 IST; 48min ago
     Main PID: 1523 (mysqld)
       Status: "Server is operational"
        Tasks: 37 (limit: 9262)
       Memory: 286.1M
          CPU: 22.434s
      CGroup: /system.slice/mysql.service
              └─1523 /usr/sbin/mysqld

Aug 03 12:48:48 apsit-HP-280-G3-MT systemd[1]: Starting MySQL Community Server...
Aug 03 12:49:28 apsit-HP-280-G3-MT systemd[1]: Started MySQL Community Server.
apsit@apsit-HP-280-G3-MT:~/Desktop/Shreya$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46-0ubuntu0.22.04.3 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| FD1      |
| information_schema |
| labdemo  |
+-----+
```

Starting mysql

```
mysql> CREATE DATABASE employeeId;
Query OK, 1 row affected (0.56 sec)

mysql> CREATE TABLE employee (
  ->     employeeId INT PRIMARY KEY,
  ->     name VARCHAR(50),
  ->     salary INT
  -> );
```

Creating database employeeId



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```
mysql> CREATE TABLE employee ( employeeId INT PRIMARY KEY, name VARCHAR(50), salary INT );
ERROR 1046 (3D000): No database selected
mysql> USE employeeId;
Database changed
mysql> CREATE TABLE employee ( employeeId INT PRIMARY KEY, name VARCHAR(50), salary INT );
Query OK, 0 rows affected (1.22 sec)

mysql> INSERT INTO employee VALUES (23104134, 'Aditi', 75000), (23104124, 'Siddhi', 60000), (23104148, 'Shreya', 80000);
Query OK, 3 rows affected (0.19 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM employee;
+-----+-----+-----+
| employeeId | name | salary |
+-----+-----+-----+
| 23104124 | Siddhi | 60000 |
| 23104134 | Aditi | 75000 |
| 23104148 | Shreya | 80000 |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

Adding all the data in table

Conclusion: In this experiment we understood the open-source database management tools such as Cassandra and MySQL. We learned how to create databases, tables, insert and retrieve data, and gained basic knowledge of the steps involved in contributing to the Cassandra open-source project.